Backpropagation

AI · University · Lecture 2



In this lecture

- Draw a network as a graph.
- Backpropagate by hand.
- Explain vanishing gradients.

Remember

- Local gradients multiply.
- One backward pass gives all gradients.
- Check gradients numerically.

Three nodes, real numbers

1. Warm-up (5 min): Who changes the weights?
2. Model (15 min): A worked example.
3. Practise (20 min): A neuron by hand.
4. Check (15 min): Finite differences.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Where did your hand gradient differ from the check?
2. Why do residual connections help?
3. What should students derive, and what can they import?

Exit ticket

- Backpropagate through $y = (2w)^2$ at $w = 1$.